

Ghanashyam D

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SKILLS

Languages: JavaScript, Python, C++, Java

Frontend: React.js, Tailwind CSS

Backend: Node.js, Express.js, Flask, Spring Boot

Databases: PostgreSQL, MongoDB, Redis, ChromaDB

Machine Learning & AI: Scikit-learn, Pandas, NumPy, Retrieval-Augmented Generation (RAG), Sentence Embeddings

Tools & Frameworks: LangChain, Hugging Face Transformers, Ollama, Docker, AWS (EC2), Git, GitHub

Concepts: REST API Design, JWT Authentication, Vector Search, Information Retrieval, System Design, Rate Limiting, Caching

PROJECTS

FraudGuard — Transaction Monitoring System

[Code](#) | [Live Demo](#)

- Designed a backend-driven system to process transaction data and detect anomalous patterns in near real-time.
- Integrated a machine learning model (Scikit-learn) to classify transactions and identify potential fraud scenarios.
- Built REST APIs using Node.js and Express for data ingestion, processing, and prediction workflows.
- Developed a React dashboard to visualize transaction activity, alerts, and risk indicators.
- Used PostgreSQL for structured data storage and Docker for containerized deployment on AWS EC2.
- Structured the system with modular components to support maintainability and future scalability.

API Rate Limiter — Traffic Control System

[Code](#) | [Live Demo](#)

- Implemented an API rate limiting system using the Token Bucket approach to control request flow per user.
- Used Redis to store and manage request tokens, enabling efficient handling of concurrent requests.
- Designed dynamic rate limits based on API keys (FREE vs PREMIUM users) with configurable thresholds.
- Built backend services using Spring Boot with interceptor-based request filtering.
- Developed a minimal React interface to demonstrate request limits and token refill behavior.
- Deployed backend on Render and frontend on Vercel using environment-based configuration.

LlamaChain — AI-Powered Multi-Document Knowledge Assistant

[Code](#)

- Built a Retrieval-Augmented Generation (RAG) system for conversational question answering, summarization, and comparison across PDF and PowerPoint documents using local LLMs.
- Implemented multiple retrieval pipelines including BM25, Dense Retrieval, Hybrid Retrieval, Query Expansion, and Reranking to evaluate retrieval effectiveness.
- Developed document chunking, sentence embedding generation, ChromaDB vector indexing, and metadata-aware retrieval workflows.
- Integrated local LLM inference using Ollama and LangChain to enable document-grounded response generation.
- Evaluated retrieval and answer quality using Hit@5, Precision@5, MRR, ROUGE-L, BLEU, BERTScore, and LLM-as-Judge metrics.

RESEARCH PUBLICATION

Impact of Advanced Retrieval Techniques on Answer Quality in Document-Grounded RAG Systems *Under Review*

- Conducted a comparative study of retrieval pipelines including BM25, Dense Retrieval, Hybrid Retrieval, Query Expansion, and Reranking within a Retrieval-Augmented Generation framework.
- Benchmarked retrieval effectiveness using Hit@5, Precision@5, and Mean Reciprocal Rank (MRR).
- Evaluated answer generation quality using ROUGE-L, BLEU, BERTScore, and LLM-as-Judge metrics.
- Analyzed the relationship between retrieval optimization techniques and downstream answer quality in document-grounded question answering systems.

EDUCATION

JSS Science and Technology University, Mysuru

May 2026

B.E. in Computer Science and Engineering — Current CGPA: 8.8/10

Vijaya Vitthala Composite PU College, Mysuru

May 2021

Senior Secondary (PUC) — 97.5%

Kautilya Vidyalaya, Mysuru

May 2019

Secondary (SSLC) — 91.2%